AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	28.8 / 4.73172 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-08 00:00 UTC
Location	29.8342, -95.4891
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 839	39.12 to 100; mean 73.707	55.64 at 2026-07-07 23:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 839	21.6667 to 36.1111; mean 28.8204	32 at 2026-07-07 23:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1094	0 to 360; mean 134.232	0 at 2026-07-07 23:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1126	0 to 15.4333; mean 2.2433	0 at 2026-07-07 23:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 130	5891.13 to 5924.31; mean 5909.1	5911.33 at 2026-07-08 00:00Z; at event time
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 130	19.6394 to 2239.24; mean 721.26	1282.97 at 2026-07-08 00:00Z; at event time
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 130	37.6598 to 52.9161; mean 45.7319	49.6827 at 2026-07-08 00:00Z; at event time
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 130	1006.52 to 1017.31; mean 1013.23	1012.12 at 2026-07-08 00:00Z; at event time
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 130	17.4065 to 20.4926; mean 19.0969	18.4321 at 2026-07-08 00:00Z; at event time
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 130	-2.83889 to 1.82714; mean 0.3481	-0.758887 at 2026-07-08 00:00Z; at event time
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 130	-0.854414 to 4.9494; mean 2.1456	3.8694 at 2026-07-08 00:00Z; at event time
openaq	Ozone (ozone)	ppm	17 / 785	-0.004 to 0.069; mean 0.0172	0.015 at 2026-07-08 00:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 150	8 to 75; mean 25.5467	41 at 2026-07-08 00:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	79 / 2718	0 to 62.1; mean 5.5521	12.6 at 2026-07-08 00:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 26.8861	43 at 2026-07-08 00:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 360	38 to 95; mean 75.7333	63 at 2026-07-08 00:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 5.62; mean 0.0476	0 at 2026-07-08 00:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 360	1012 to 1019; mean 1016.26	1014 at 2026-07-08 00:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 360	23.78 to 36.05; mean 29.5481	32.03 at 2026-07-08 00:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 359; mean 190.558	111 at 2026-07-08 00:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.36 to 8.39; mean 2.2549	0.89 at 2026-07-08 00:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	74 / 5125	0 to 146.4; mean 7.3305	18.7 at 2026-07-08 00:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0243723 to 0.0253077; mean 0.0249	0.0249873 at 2026-07-07 18:57Z; 5 h 2 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	8 / 8	0.000174097 to 0.000235341; mean 0.0002	0.000202795 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	3.63106e-05 to 4.30727e-05; mean 0	4.10314e-05 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	8 / 8	-3.49304e-05 to 4.72775e-05; mean 0	2.51087e-05 at 2026-07-07 20:37Z; 3 h 22 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; 3 h 22 min before event
tceq	Carbon monoxide (co)	ppm	3 / 162	0 to 1; mean 0.237	0.6 at 2026-07-08 00:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 637	0 to 35.8; mean 6.5196	28.8 at 2026-07-08 00:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 154	-0.5 to 6.8; mean -0.0481	-0.4 at 2026-07-08 00:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-06 12Z 70.19, 18Z 55.49, 07-07 00Z 77.29, 06Z 89.88, 12Z 71, 18Z 57, 07-08 00Z 79.08, 06Z 92.24, 12Z 70.83, 18Z 59.59, 07-09 00Z 77.1, 06Z 89.2
asos	Air temperature (temperature)	07-06 12Z 30.32, 18Z 32.62, 07-07 00Z 28.31, 06Z 25.72, 12Z 29.83, 18Z 32.19, 07-08 00Z 25.82, 06Z 24.3, 12Z 30.32, 18Z 31.65, 07-09 00Z 28.27, 06Z 25.68
asos	Wind direction (wind_direction)	07-06 12Z 180.9, 18Z 166.7, 07-07 00Z 137.1, 06Z 73.4, 12Z 133, 18Z 148.1, 07-08 00Z 105.8, 06Z 27.45, 12Z 187.1, 18Z 171.8, 07-09 00Z 158.6, 06Z 126.2
asos	Wind speed (wind_speed)	07-06 12Z 2.058, 18Z 3.818, 07-07 00Z 2.083, 06Z 0.7665, 12Z 1.424, 18Z 3.304, 07-08 00Z 2.915, 06Z 0.3517, 12Z 2.173, 18Z 3.988, 07-09 00Z 2.757, 06Z 1.442
noaa_gfs	500-hPa geopotential height (gh_500)	07-06 12Z 5892, 18Z 5902, 07-07 00Z 5898, 06Z 5905, 12Z 5899, 18Z 5914, 07-08 00Z 5912, 06Z 5918, 12Z 5911, 18Z 5924, 07-09 00Z 5915, 06Z 5919, 12Z 5910
noaa_gfs	Planetary boundary layer height (pbl_height)	07-06 12Z 175.4, 18Z 1810, 07-07 00Z 1019, 06Z 227.9, 12Z 126.9, 18Z 1996, 07-08 00Z 976.4, 06Z 250.5, 12Z 150.5, 18Z 1936, 07-09 00Z 244.1, 06Z 303.7, 12Z 159.8
noaa_gfs	Precipitable water (precipitable_water)	07-06 12Z 41.89, 18Z 42.78, 07-07 00Z 46.68, 06Z 45.56, 12Z 46.22, 18Z 48.1, 07-08 00Z 48.87, 06Z 47.17, 12Z 46.64, 18Z 48.04, 07-09 00Z 45.44, 06Z 43.81, 12Z 43.31
noaa_gfs	Surface pressure (surface_pressure)	07-06 12Z 1012, 18Z 1012, 07-07 00Z 1010, 06Z 1013, 12Z 1014, 18Z 1014, 07-08 00Z 1012, 06Z 1015, 12Z 1015, 18Z 1015, 07-09 00Z 1013, 06Z 1014, 12Z 1014
noaa_gfs	850-hPa temperature (t_850)	07-06 12Z 19.67, 18Z 17.97, 07-07 00Z 19.75, 06Z 19.78, 12Z 18.77, 18Z 17.85, 07-08 00Z 19.03, 06Z 19.77, 12Z 19.06, 18Z 17.99, 07-09 00Z 19.87, 06Z 19.53, 12Z 19.2
noaa_gfs	10-m eastward wind (u_10m)	07-06 12Z 1.021, 18Z 0.7598, 07-07 00Z -0.6914, 06Z 0.346, 12Z 0.8611, 18Z -0.1427, 07-08 00Z -1.34, 06Z 0.2456, 12Z 0.8352, 18Z 0.9613, 07-09 00Z -0.07786, 06Z 0.7396, 12Z 1.007
noaa_gfs	10-m northward wind (v_10m)	07-06 12Z 1.384, 18Z 1.541, 07-07 00Z 3.52, 06Z 2.189, 12Z 1.199, 18Z 1.606, 07-08 00Z 4.025, 06Z 2.566, 12Z 1.287, 18Z 1.112, 07-09 00Z 3.109, 06Z 2.865, 12Z 1.49
openaq	Ozone (ozone)	07-06 12Z 0.01262, 18Z 0.03414, 07-07 00Z 0.01345, 06Z 0.005189, 12Z 0.01275, 18Z 0.03023, 07-08 00Z 0.02554, 06Z 0.004395, 12Z 0.004094, 18Z 0.03667
openaq	PM10 (pm10)	07-06 12Z 30.22, 18Z 27.5, 07-07 00Z 15.89, 06Z 18.28, 12Z 33.17, 18Z 29.88, 07-08 00Z 21.94, 06Z 24.39, 12Z 36, 18Z 23
openaq	PM2.5 (pm25)	07-06 12Z 6.452, 18Z 5.623, 07-07 00Z 4.88, 06Z 6.191, 12Z 6.682, 18Z 7.206, 07-08 00Z 4.586, 06Z 6.593, 12Z 5.806, 18Z 3.239, 07-09 00Z 3.701, 06Z 4.591, 12Z 4.524
openweather	Cloud cover (cloud_cover)	07-06 12Z 24.47, 18Z 24.9, 07-07 00Z 14.61, 06Z 4.033, 12Z 36.77, 18Z 46.57, 07-08 00Z 27.8, 06Z 4.667, 12Z 6.655, 18Z 29.1, 07-09 00Z 69.65, 06Z 31.8
openweather	Relative humidity (humidity)	07-06 12Z 78.2, 18Z 59.53, 07-07 00Z 75.58, 06Z 87.8, 12Z 77.07, 18Z 59.23, 07-08 00Z 76.6, 06Z 90.5, 12Z 79.59, 18Z 60.79, 07-09 00Z 75.35, 06Z 88.2
openweather	Precipitation rate (precipitation)	07-06 12Z 0, 18Z 0.005333, 07-07 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-08 00Z 0.5237, 06Z 0, 12Z 0, 18Z 0.04379, 07-09 00Z 0, 06Z 0
openweather	Surface pressure (pressure)	07-06 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016, 12Z 1018, 18Z 1016, 07-08 00Z 1016, 06Z 1017, 12Z 1019, 18Z 1017, 07-09 00Z 1016, 06Z 1016
openweather	Air temperature (temperature)	07-06 12Z 29.63, 18Z 34.39, 07-07 00Z 29.5, 06Z 26.39, 12Z 29.93, 18Z 34.09, 07-08 00Z 27.7, 06Z 24.53, 12Z 29.55, 18Z 33.46, 07-09 00Z 29.26, 06Z 26.28
openweather	Wind direction (wind_direction)	07-06 12Z 233.5, 18Z 169.2, 07-07 00Z 178.6, 06Z 178, 12Z 214.6, 18Z 175.7, 07-08 00Z 170.7, 06Z 193.8, 12Z 233.3, 18Z 190.1, 07-09 00Z 178.8, 06Z 172.6
openweather	Wind speed (wind_speed)	07-06 12Z 1.838, 18Z 2.215, 07-07 00Z 2.186, 06Z 1.73, 12Z 1.439, 18Z 2.646, 07-08 00Z 3.145, 06Z 2.234, 12Z 2.152, 18Z 2.746, 07-09 00Z 2.792, 06Z 1.932

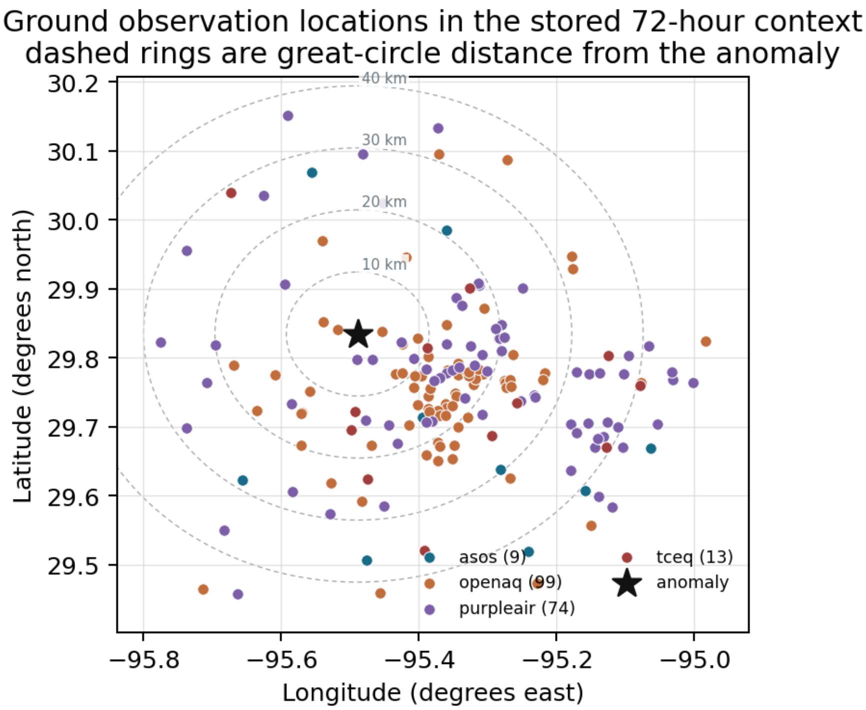
Source	Metric	Values
purpleair	PM2.5 (pm25)	07-06 12Z 8.371, 18Z 6.384, 07-07 00Z 6.413, 06Z 7.879, 12Z 7.973, 18Z 8.328, 07-08 00Z 7.524, 06Z 10, 12Z 7.866, 18Z 5.304, 07-09 00Z 5.023, 06Z 6.842, 12Z 8.4
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	CO column (s5p_co_column)	07-06 18Z 0.02528, 07-07 18Z 0.02495, 07-08 18Z 0.02437
sentinel5p	CO granules available (s5p_co_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-06 18Z 0.0002254, 07-07 18Z 0.0002152, 07-08 18Z 0.000185
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-06 18Z 4.086e-05, 07-07 18Z 3.867e-05, 07-08 18Z 4.307e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-06 18Z 4.559e-05, 07-07 18Z 2.628e-06, 07-08 18Z -2.084e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
tceq	Carbon monoxide (co)	07-06 12Z 0.2353, 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.25, 12Z 0.2167, 18Z 0.2706, 07-08 00Z 0.2, 06Z 0.2444, 12Z 0.25, 07-09 06Z 0.2222, 12Z 0.5
tceq	Nitrogen dioxide (no2)	07-06 12Z 5.182, 18Z 4.274, 07-07 00Z 4, 06Z 6.039, 12Z 5.905, 18Z 7.493, 07-08 00Z 6.903, 06Z 10.96, 12Z 7.344, 18Z 1.1, 07-09 06Z 5.981, 12Z 9.275
tceq	Sulfur dioxide (so2)	07-06 12Z 0.1353, 18Z 0.075, 07-07 00Z -0.23, 06Z -0.2687, 12Z 0.3333, 18Z -0.05294, 07-08 00Z -0.2, 06Z -0.1867, 12Z -0.03889, 07-09 06Z -0.1556, 12Z -0.1

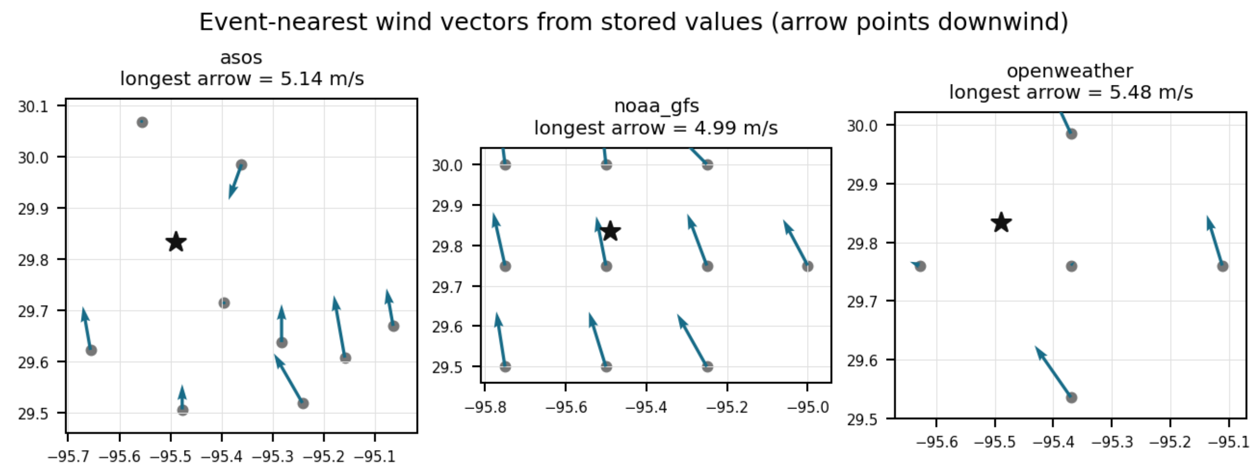
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (16.161 km) 72.53, IAH (20.788 km) 72.76, DWH (26.788 km) 79.7, SGR (28.573 km) 76.96, HOU (29.611 km) 72.62, AXH (36.503 km) 73.08, EFD (40.678 km) 68.57, LVJ (42.433 km) 71.95, +1 more
asos	Air temperature (temperature)	MCJ (16.161 km) 29.01, IAH (20.788 km) 28.67, DWH (26.788 km) 27.11, SGR (28.573 km) 28.72, HOU (29.611 km) 29.39, AXH (36.503 km) 28.32, EFD (40.678 km) 30.37, LVJ (42.433 km) 29.18, +1 more
asos	Wind direction (wind_direction)	MCJ (16.161 km) 142.9, IAH (20.788 km) 157.4, DWH (26.788 km) 85.88, SGR (28.573 km) 152.6, HOU (29.611 km) 165.4, AXH (36.503 km) 78.93, EFD (40.678 km) 145.6, LVJ (42.433 km) 112.8, +1 more
asos	Wind speed (wind_speed)	MCJ (16.161 km) 2.909, IAH (20.788 km) 2.392, DWH (26.788 km) 1.252, SGR (28.573 km) 2.572, HOU (29.611 km) 2.855, AXH (36.503 km) 0.915, EFD (40.678 km) 3.195, LVJ (42.433 km) 1.779, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.50 (9.422 km) 5909, gfs:30.00,-95.50 (18.465 km) 5909, gfs:29.75,-95.25 (24.902 km) 5909, gfs:29.75,-95.75 (26.859 km) 5909, gfs:30.00,-95.25 (29.512 km) 5909, gfs:30.00,-95.75 (31.178 km) 5909, gfs:29.50,-95.50 (37.177 km) 5910, gfs:29.50,-95.25 (43.758 km) 5910, +2 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.50 (9.422 km) 812.8, gfs:30.00,-95.50 (18.465 km) 921.9, gfs:29.75,-95.25 (24.902 km) 821, gfs:29.75,-95.75 (26.859 km) 620.4, gfs:30.00,-95.25 (29.512 km) 897.2, gfs:30.00,-95.75 (31.178 km) 579.5, gfs:29.50,-95.50 (37.177 km) 636.2, gfs:29.50,-95.25 (43.758 km) 588.5, +2 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.50 (9.422 km) 45.57, gfs:30.00,-95.50 (18.465 km) 45.9, gfs:29.75,-95.25 (24.902 km) 46.51, gfs:29.75,-95.75 (26.859 km) 44.79, gfs:30.00,-95.25 (29.512 km) 47.24, gfs:30.00,-95.75 (31.178 km) 44.89, gfs:29.50,-95.50 (37.177 km) 44.76, gfs:29.50,-95.25 (43.758 km) 45.74, +2 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.50 (9.422 km) 1013, gfs:30.00,-95.50 (18.465 km) 1012, gfs:29.75,-95.25 (24.902 km) 1015, gfs:29.75,-95.75 (26.859 km) 1012, gfs:30.00,-95.25 (29.512 km) 1013, gfs:30.00,-95.75 (31.178 km) 1010, gfs:29.50,-95.50 (37.177 km) 1014, gfs:29.50,-95.25 (43.758 km) 1015, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.50 (9.422 km) 19.09, gfs:30.00,-95.50 (18.465 km) 19.08, gfs:29.75,-95.25 (24.902 km) 19.03, gfs:29.75,-95.75 (26.859 km) 19.15, gfs:30.00,-95.25 (29.512 km) 19.02, gfs:30.00,-95.75 (31.178 km) 19.21, gfs:29.50,-95.50 (37.177 km) 19.09, gfs:29.50,-95.25 (43.758 km) 19.11, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.50 (9.422 km) 0.3685, gfs:30.00,-95.50 (18.465 km) 0.6292, gfs:29.75,-95.25 (24.902 km) 0.3454, gfs:29.75,-95.75 (26.859 km) 0.5731, gfs:30.00,-95.25 (29.512 km) -0.09693, gfs:30.00,-95.75 (31.178 km) 0.8569, gfs:29.50,-95.50 (37.177 km) 0.1892, gfs:29.50,-95.25 (43.758 km) 0.2238, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.50 (9.422 km) 2.288, gfs:30.00,-95.50 (18.465 km) 2.146, gfs:29.75,-95.25 (24.902 km) 2.182, gfs:29.75,-95.75 (26.859 km) 2.46, gfs:30.00,-95.25 (29.512 km) 1.93, gfs:30.00,-95.75 (31.178 km) 2.434, gfs:29.50,-95.50 (37.177 km) 1.995, gfs:29.50,-95.25 (43.758 km) 1.907, +2 more
openaq	Ozone (ozone)	319 (0.008 km) 0.0156, 1319333 (11.294 km) 0.02115, 311 (15.421 km) 0.01672, 310 (17.39 km) 0.02088, 287 (18.775 km) 0.01842, 2046 (19.778 km) 0.01744, 312 (23.429 km) 0.01646, 325 (24.945 km) 0.0056, +9 more
openaq	PM10 (pm10)	15852419 (18.045 km) 27.5, 13380082 (24.981 km) 33, 271 (39.309 km) 16.47
openaq	PM2.5 (pm25)	10762809 (2.905 km) 9.653, 12337668 (3.368 km) 7.042, 14140750 (5.232 km) 6.688, 8967015 (6.377 km) 5.567, 8967363 (8.26 km) 7.353, 13787197 (8.381 km) 7.015, 8967413 (8.848 km) 6.67, 8967373 (10.454 km) 7.197, +71 more
openweather	Cloud cover (cloud_cover)	grid:center (14.139 km) 26.94, grid:west (15.756 km) 22.53, grid:north (20.333 km) 37.36, grid:south (35.126 km) 24.22, grid:east (37.388 km) 23.38
openweather	Relative humidity (humidity)	grid:center (14.139 km) 76.19, grid:west (15.756 km) 75.65, grid:north (20.333 km) 73.82, grid:south (35.126 km) 74.21, grid:east (37.388 km) 78.79
openweather	Precipitation rate (precipitation)	grid:center (14.139 km) 0.08208, grid:west (15.756 km) 0.02028, grid:north (20.333 km) 0.01014, grid:south (35.126 km) 0.001944, grid:east (37.388 km) 0.1236
openweather	Surface pressure (pressure)	grid:center (14.139 km) 1016, grid:west (15.756 km) 1016, grid:north (20.333 km) 1016, grid:south (35.126 km) 1016, grid:east (37.388 km) 1016
openweather	Air temperature (temperature)	grid:center (14.139 km) 29.72, grid:west (15.756 km) 29.49, grid:north (20.333 km) 29.47, grid:south (35.126 km) 29.51, grid:east (37.388 km) 29.56
openweather	Wind direction (wind_direction)	grid:center (14.139 km) 205.3, grid:west (15.756 km) 177.4, grid:north (20.333 km) 173.9, grid:south (35.126 km) 201.8, grid:east (37.388 km) 194.5
openweather	Wind speed (wind_speed)	grid:center (14.139 km) 2.053, grid:west (15.756 km) 1.977, grid:north (20.333 km) 2.208, grid:south (35.126 km) 2.16, grid:east (37.388 km) 2.876
purpleair	PM2.5 (pm25)	51959 (4.098 km) 6.977, 301969 (4.533 km) 10.2, 155367 (6.247 km) 8.124, 223813 (8.787 km) 7.406, 302037 (11.11 km) 9.547, 246011 (12.486 km) 6.386, 130763 (12.96 km) 2.36, 296101 (13.109 km) 7.91, +66 more
tceq	Carbon monoxide (co)	482011052 (10.018 km) 0.3893, 482011035 (24.981 km) 0.1216, 482011039 (39.322 km) 0.1891
tceq	Nitrogen dioxide (no2)	482010047 (0.0 km) 7.938, 482011052 (10.018 km) 13.83, 482011066 (12.524 km) 5.505, 482010055 (15.427 km) 2.99, 482010024 (17.385 km) 6.465, 482010416 (24.952 km) 4.863, 482011035 (24.981 km) 10.48, 482010029 (28.955 km) 2.558, +4 more
tceq	Sulfur dioxide (so2)	482010051 (23.42 km) -0.1241, 482011035 (24.981 km) -0.05385, 482011039 (39.322 km) 0.04375


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
10 km
20 km
30 km
40 km
Latitude (degrees north)
Longitude (degrees east)
asos (9)
openaq (99)
purpleair (74)
tceq (13)
anomaly


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 5.14 m/s
noaa_gfs
longest arrow = 4.99 m/s
openweather
longest arrow = 5.48 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

Surface wind observations from ASOS stations recorded calm conditions reaching 0.0 m/s near the anomaly location around 2026-07-08T00:00:00Z, facilitating localized pollutant accumulation through atmospheric stagnation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

Regional smoke transport or a broad sulfur-rich industrial episode is not strongly supported because Sentinel-5P CO and NO2 columns near Houston were not unusually elevated before the event, Sentinel-5P SO2 was near zero on 2026-07-07 18Z, TCEQ SO2 near the event was -0.4 ppb, and area PM2.5 remained modest at 4.586 ug/m3 in the 00Z six-hour mean.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The TCEQ monitoring network recorded a severe NO2 concentration anomaly reaching 28.8 ppb at 2026-07-08T00:00:00Z in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

The air quality anomaly in Houston, Texas on July 8th was characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

ASOS meteorological stations near the anomaly site observed calm surface winds with speeds dropping to 0.0 m/s around 2026-07-07T23:55:00Z.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

The anomalous TCEQ NO2 value of 28.8 ppb exceeds the expected value of 4.731724137931035 ppb by about 24.1 ppb and has a reported z-score of 6.803119928227577 with severe severity.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

The air quality anomaly in Houston, Texas is not caused by high levels of NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The Houston NO2 anomaly is consistent with enhanced nighttime accumulation because the nearest ASOS station reported calm winds at 23:55Z, the 06Z six-hour ASOS wind-speed mean was only 0.3517 m/s across the area, and GFS planetary boundary layer height dropped from about 1283 m at 00Z to about 251 m by 06Z, indicating increasingly poor dispersion overnight.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

Weak near-surface dispersion is a likely contributing cause because the nearest ASOS station reported calm wind at 23:55Z, the 06Z ASOS 6-hour mean wind speed dropped to 0.35 m/s, and very light winds favor local NO2 buildup near sources in the Houston urban boundary layer.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

A short-duration plume from nearby combustion sources such as roadway traffic, industrial combustion, or diesel equipment is a likely physical cause because the anomalous TCEQ monitor measured 28.8 ppb NO2 at 00Z while nearby area-wide TCEQ NO2 conditions were much lower on average and the closest TCEQ CO monitor showed contemporaneous combustion-related CO present at 0.6 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

A severe thunderstorm or tornado event in the region has stirred up dust and pollutants, contributing to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

TCEQ ground monitors detected a sharp increase in surface NO2 to 28.8 ppb at the anomaly site, aligned with an increase in total NO2 column density observed by Sentinel-5P over the larger Houston domain.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The pollutant anomaly is a TCEQ NO2 measurement of 28.8 ppb at 2026-07-08T00:00:00+00:00 at coordinates 29.834, -95.489 in Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The air quality anomaly is related to PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

The Houston NO2 anomaly is consistent with a short-lived nearby combustion plume because the anomalous TCEQ monitor measured 28.8 ppb NO2 at 00Z while the nearest colocated ozone observation was only 0.015 ppm and nearby TCEQ CO reached 0.6 ppm, which is a fresh-emission pattern with titration of ozone by recently emitted NO.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

Accumulated local evening rush-hour vehicular emissions under calm atmospheric conditions directly contributed to the sharp spike in NO2 concentrations observed by TCEQ.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

The TCEQ monitor at 29.834, -95.489 recorded 28.8 ppb NO2 at 2026-07-08T00:00:00Z, which was far above the monitor-specific expected value of 4.73 ppb and above the contemporaneous 6-hour area mean of 6.90 ppb, indicating a strongly localized NO2 enhancement.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

A severe nitrogen dioxide (NO_2) concentration anomaly of 28.8 ppb was detected at coordinates (29.834, -95.489) on 2026-07-08T00:00:00+00:00, significantly exceeding the expected baseline value of 4.73 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

NOAA GFS model data shows planetary boundary layer height collapsing from nearly 2000 meters at 18:00 UTC on July 7 to 976 meters by 00:00 UTC on July 8, 2026, trapping ground-level nitrogen dioxide emissions within a shrinking boundary layer.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

The anomaly occurred on July 8th, 2026, around 00:01:52 UTC, and was located approximately 15.756 km from the query point.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

The PM2.5 levels are elevated at 18.7 ug/m3, which is higher than the mean of 7.33 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

The anomaly was likely caused by a combination of factors, including local emissions and weather patterns.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Sentinel-5P recorded an elevated tropospheric NO2 column density averaging 4.31e-5 mol/m^2 over the area during the 18Z window on July 8, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

The air quality anomaly in Houston, Texas is not caused by high levels of SO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

Fresh local NOx emissions with limited immediate oxidation are a likely cause because ozone near the anomaly was only 0.015 ppm at 00Z, TCEQ SO2 near the time was not elevated, PM2.5 remained modest across OpenAQ and PurpleAir, and Sentinel-5P did not indicate a coincident regional combustion or sulfur event over Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

The 6-hour mean across 12 TCEQ NO2 monitors for 2026-07-08 00Z is 6.903 ppb, which is far below the anomalous monitor reading of 28.8 ppb at the same time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

NO2 levels were also significantly higher than normal during the same period.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

NOAA GFS model output showed the planetary boundary layer height rapidly dropping from nearly 2000 meters at 18:00 UTC down to approximately 976 meters at 00:00 UTC on July 8.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

Near-zero surface wind speeds measured by ASOS stations hindered horizontal ventilation, preventing the dispersion of emitted nitrogen dioxide near the TCEQ station.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

OpenAQ ozone near the anomaly location was 0.015 ppm at 2026-07-08T00:00:00Z, which is consistent with a fresh NOx plume that had not produced elevated ozone and may have locally suppressed ozone through titration.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to the accumulation of pollutants in the area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The collapse of the planetary boundary layer height depicted in GFS reanalysis constrained vertical mixing, trapping urban NOx emissions close to the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

Surface wind speed measured near the anomaly location dropped to 0.0 m/s at 2026-07-07T23:55:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

ASOS observations nearest the anomaly showed 0.0 m/s wind at 2026-07-07T23:55:00Z and the 06Z period had a 6-hour mean wind speed of only 0.35 m/s, supporting weak dispersion that would trap a nearby combustion plume near the surface.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.89 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.38737025451646634 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.